

# BUDDAREDDY DIVIJ

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## Professional Summary

B.Tech graduate in Artificial Intelligence and Machine Learning with hands-on experience developing software applications and machine learning solutions using Python. Experienced in application development, data processing, testing, debugging, model development, and deployment through internship and academic projects. Strong analytical and problem-solving skills with a focus on building reliable and efficient software solutions.

## Education

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| <b>Kalasalingam Academy of Research and Education, Tamil Nadu</b>    | 2022 – 2026 |
| Bachelor of Technology in Artificial Intelligence & Machine Learning | CGPA: 7.8   |
| <b>Sri Chaitanya Junior College, Vijayawada</b>                      | 87%         |
| <b>Dr. KKR Gowtham School, Vijayawada</b>                            | 100%        |

## Experience

**Software and Application Developer** *AI Automation Labs*  
May 2026 – Present

- Developed software and machine learning applications using Python, including data preprocessing, feature engineering, model development, testing, validation, and deployment.
- Contributed to end-to-end application development by implementing, testing, debugging, and improving Python-based solutions for business-focused use cases.

## Technical Skills

**Programming Languages:** Python, SQL

**Software Development:** Flask, Application Development, Git, GitHub

**AI & Machine Learning:** Machine Learning, Deep Learning, Natural Language Processing, Predictive Modeling, Computer Vision

**Libraries:** PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib

**Development Practices:** Testing, Debugging, Data Preprocessing, Feature Engineering, Model Evaluation, Problem Solving

**Tools & Platforms:** VS Code, Jupyter Notebook, Hugging Face

## Projects

**AI-Powered Robotic Arm** *Python, C++, Computer Vision*

- Developed an AI-powered robotic arm application for real-time color-based object detection and automated pick-and-place operations.
- Implemented computer vision-based object detection and classification for red, green, and blue objects using live camera input.

**HealthAI Platform** *HTML5, JavaScript, Python, Google Gemini, RAG, Machine Learning*

- Developed a web-based Clinical Decision Support System (CDSS) integrating a custom probabilistic Machine Learning classifier, Retrieval-Augmented Generation (RAG), and Google Gemini for symptom analysis and personalized health guidance.
- Implemented cosine similarity-based disease prediction, emergency red-flag detection, drug-allergy conflict checking, interactive 2D anatomical body-map symptom input, and automated clinical PDF report generation.

**Travel Companion** *Python, Flask, LangChain, Ollama, FAISS*

- Developed a Flask-based travel recommendation application using retrieval-augmented generation to provide destination recommendations from structured datasets.
- Implemented semantic search using FAISS and Ollama embeddings to retrieve relevant travel information and improve recommendation accuracy.

**Demand Forecasting System** *Python, Scikit-learn, Pandas, NumPy, Gradio*

- Developed a machine learning application to forecast weekly sales using historical sales data and business indicators from the Walmart dataset.
- Engineered lag and rolling-average features, trained a Random Forest Regressor achieving an  $R^2$  score of 0.9775, and deployed the application on Hugging Face Spaces.

## Certifications

- Python – HackerRank
- AI Agents Fundamentals – Hugging Face